3446425310349300626622014604312051101093282396249251196897
8283306192150828270814393659987326849047994166839657747890
2124562796195600187060805768778947870098610692265944872693
4100008726998763399003025591685820639734851035629676461160
0225159200113722741273318074829547248192807653266407023083
2754286312646671501355905966429773337131834654748547607012
4233012872135321237328732721874825264039911049700172147564
7004992922645864352265011199999999999999999999999999999999
9999999999999999999999999999999999999999999999999999999999
9999999999999999999999999999999999999999999999999999999999
99999999999999999999—about $10^{1878}$—different out-of-order page arrangements.[6] If you throw the pages in the air and then gather them in a neat stack, it is almost certain that they will wind up out of numerical order, because such configurations have enormously higher entropy— there are many more ways to achieve an out-of-order outcome —than the sole arrangement in which they are in correct numerical order.

In principle, we could use the laws of classical physics to figure out exactly where each page will land after the whole stack has been thrown in the air. So, again in principle, we could precisely predict the resulting arrangement of the pages[7] and hence (unlike in quantum mechanics, which we ignore until the next chapter) there would seem to be no need to rely on probabilistic notions such as which outcome is more or less likely than another. But statistical reasoning is both powerful and useful. If *War and Peace* were a pamphlet of only a couple of pages we just might be able to successfully complete the necessary calculations, but it would be impossible to do this for the real War and Peace.[8] Following the precise motion of 693 floppy pieces of paper as they get caught by gentle air currents and rub, slide, and flap against one another would be a monumental task, well beyond the capacity of even the most powerful supercomputer.

Moreover—and this is critical—having the exact answer wouldn't even be that useful. When you examine the resulting stack of pages, you are far less interested in the exact details of which page happens to be where than you are in the general question of whether the pages are in the correct order. If they are, great. You could sit down and continue reading about Anna Pavlovna and Nikolai Ilych Rostov, as usual. But if